

Automatic Alignment and Normalization of Lung Region Through Landmark Prediction and Warping for Automatic Classification of Pneumonia and COVID-19 with ResNet-18

Rafael Alejandro Cruz Ovando ¹, Salvador Eugenio Ayala Raggi ^{1,*} and Aldrin Barreto Flores ¹

¹ Facultad de Ciencias de la Electrónica, Benemérita Universidad Autónoma de Puebla, Puebla, México

* Correspondence: rafaelcruz.ovando@gmail.com

Abstract

This paper presents an automatic classification system for pulmonary diseases in chest X-rays based on geometric normalization. The proposed method consists of three main stages. **First stage:** ResNet-18 convolutional neural networks with coordinate attention mechanism are trained to predict 15 landmarks defining the lung contour, achieving a mean error of 3.61 pixels (median 3.07 pixels) through an ensemble of four models with test-time augmentation. **Second stage:** the predicted coordinates are used to geometrically normalize each image through Generalized Procrustes Analysis (to obtain a standard lung shape), Delaunay triangulation (to build a deformation mesh), and piecewise affine transformation (warping), eliminating variations in position, scale, and orientation. **Third stage:** normalized images are classified into three categories (COVID-19, Viral Pneumonia, and Normal) using a ResNet-18 classifier with transfer learning and SAHS preprocessing. The system was evaluated on the COVID-19 Radiography Database with 15,153 images, achieving 98.10% accuracy, 97.17% F1-Macro, and 98.09% F1-Weighted on the test set. Five-fold cross-validation on the training and validation sets demonstrated high stability ($98.60 \pm 0.26\%$ accuracy), confirming robustness of the approach. Controlled comparison between configurations demonstrates that geometric normalization achieves performance comparable to original images (0.58 percentage points difference), while simple border cropping without normalization produces inferior results. These results evidence that the proposed system learns genuine pathological features from the lung region.

Keywords: Geometric normalization; Landmark detection; COVID-19; Chest X-ray; Deep learning; Generalized Procrustes Analysis

1. Introduction

Pneumonia represents one of the leading causes of morbidity and mortality worldwide, being especially critical during the COVID-19 pandemic. Timely and accurate diagnosis of this disease is fundamental for proper patient treatment. Chest X-rays constitute a first-line diagnostic tool due to their wide availability, low cost, and rapid acquisition [1]. However, manual interpretation of these images requires specialized experience and is subject to inter-observer variability, which motivates the development of automatic classification methods.

In recent years, deep learning-based methods have demonstrated great potential for automatic pneumonia detection in radiographs [2–4]. These approaches use convolutional neural networks (CNNs) to extract discriminant features directly from images. However,

Received:

Revised:

Accepted:

Published:

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their performance can be affected by variations in image acquisition, including differences in patient position, scale, and orientation, as well as artifacts from each radiographic equipment. This phenomenon, known as *domain shift* [5], can compromise the generalization capability of models.

1.1. Related Work

Various works have addressed the variability problem in medical images through normalization and alignment techniques. Ayala-Raggi et al. [6] demonstrated that chest X-ray normalization, combined with discriminant feature selection, significantly improves automatic COVID-19 recognition. In their study presented at ACPR 2023, the authors showed that eliminating non-pathological variations allows classifiers to focus on intrinsic disease characteristics, resulting in higher diagnostic accuracy.

Complementarily, Picazo-Castillo et al. [7] conducted a comparative study of different lung image representations for automatic pneumonia recognition. Their results evidence that the way visual information is presented to the classifier has a significant impact on performance, highlighting the importance of adequately preprocessing images to facilitate learning of pathological patterns.

Other works have explored the use of geometric transformations in medical images. Spatial Transformer Networks [8] allow learning spatial transformations in an end-to-end manner, but require large amounts of data for training. In contrast, methods based on anatomical landmarks offer a more interpretable alternative and can work with smaller datasets, as they incorporate a priori knowledge about anatomy.

1.2. Proposal of the Present Work

In this work, we propose a geometric normalization method that goes beyond traditional rigid transformations (rotation, translation, scaling). The proposed system incorporates local deformations based on lung anatomy to transform each radiograph to a standard shape, eliminating not only global pose variations but also differences in thorax morphology between patients.

The method is based on three main components. First, convolutional neural networks are used to automatically predict the coordinates of 15 landmarks defining the bilateral lung contour. Second, Generalized Procrustes Analysis [9] is applied to a set of manually annotated landmarks to determine a standard lung shape representative of the dataset. Third, Delaunay triangulation [10] is applied to build a triangle mesh over the landmarks, and piecewise affine warping [11] is performed to deform each image to match its predicted landmarks with the standard shape. The resulting normalized images are used to train a ResNet-18 classifier [12] with transfer learning.

This approach allows obtaining a standardized representation of the lung region where non-pathological variability (due to anatomical, position, or acquisition differences) has been minimized, allowing the classifier to focus on learning the intrinsic characteristics of each pathology. Notably, this work represents the first application of piecewise affine warping based on automatically detected anatomical landmarks for COVID-19 classification, combining classical shape analysis (GPA) with modern deep learning in a fully automatic and interpretable pipeline.

2. Materials and Methods

2.1. System Overview

From a high-level perspective, the proposed system can be understood as a “black box” that receives a chest X-ray as input and produces a diagnosis (Normal, COVID-19, or Viral Pneumonia) as output. Internally, the system executes three processing stages:

(1) prediction of anatomical landmarks, (2) geometric normalization of the image, and (3) classification of the normalized image. This modular architecture allows each component to be trained and evaluated independently.

Figure 1 illustrates the complete processing flow. An input image first passes through the landmark prediction model, which generates 15 pairs of (x, y) coordinates describing the lung contour. These coordinates, together with the pre-calculated standard shape, are used to perform the geometric transformation. The resulting image, where the lung region has been normalized to a standard shape, is finally processed by the classifier that emits the diagnosis.

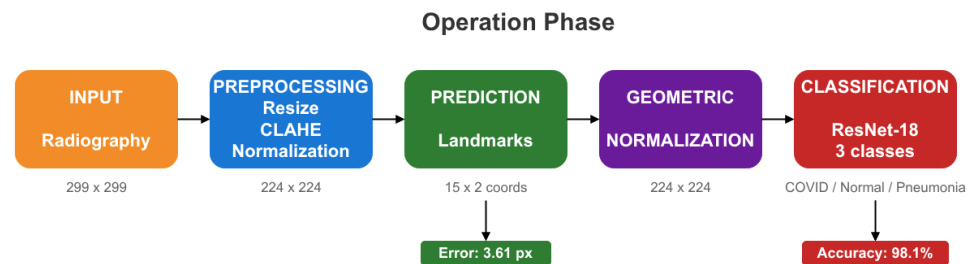


Figure 1. Operation flow of the proposed system. From left to right: (1) radiograph input (299×299), (2) preprocessing with resizing, CLAHE, and normalization ($224 \times 224 \times 3$), (3) prediction of 15 coordinate pairs with 3.61 px error, (4) geometric normalization through warping ($224 \times 224 \times 3$), and (5) classification with ResNet-18 into three categories (COVID-19, Normal, Viral Pneumonia).

2.2. Dataset

The public **COVID-19 Radiography Database** [4,17] was used, containing 15,153 posteroanterior radiographs organized into three categories: COVID-19 (3,616 images), Normal (10,192 images), and Viral Pneumonia (1,345 images). Original images (299×299 pixels) were resized to 224×224 pixels, standard format for ImageNet-pretrained models.

For supervised training of the landmark prediction model, 15 landmarks were manually annotated on the lung contour of 957 images. Landmarks define control points on the bilateral lung silhouette, organized in a central vertical axis (5 points) and five symmetric left/right pairs (10 points). The dataset was split in a stratified manner into training (75%), validation (15%), and test (10%).

2.3. Landmark Prediction Model Architecture

The prediction model is based on a ResNet-18 [12] architecture pretrained on ImageNet [13]. This transfer learning approach leverages visual features learned from millions of images (edges, textures, patterns) as a basis for the specific landmark localization task, which is particularly beneficial given the limited size of the annotated dataset.

The architecture is structured in three main components (Figure 2):

- **Backbone:** ResNet-18 convolutional layers process the input image ($224 \times 224 \times 3$) producing a $7 \times 7 \times 512$ feature map.
- **Coordinate Attention Module:** Inserted between the backbone and regression head, this mechanism [14] captures positional dependencies along both spatial dimensions, crucial for precise localization tasks.
- **Regression Head:** Three fully connected layers ($512 \rightarrow 512 \rightarrow 768 \rightarrow 30$) with Group Normalization [15] and staggered dropout. The output is 30 normalized values in $[0,1]$, corresponding to the (x, y) coordinates of the 15 landmarks.

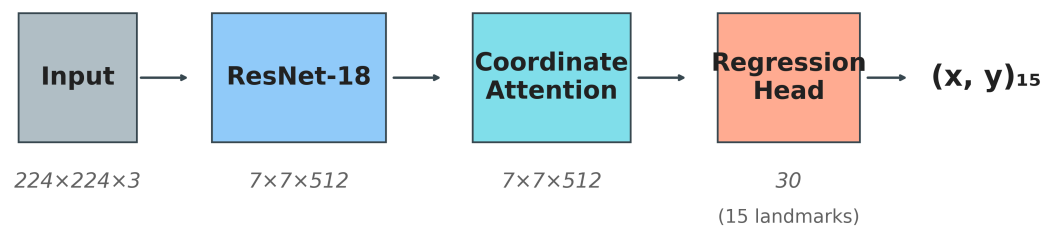


Figure 2. Landmark prediction model architecture. The input image ($224 \times 224 \times 3$) passes through the ResNet-18 backbone that produces a feature map ($7 \times 7 \times 512$), then through the Coordinate Attention module that maintains dimensions, and finally through the regression head that produces 30 values corresponding to the (x, y) coordinates of the 15 landmarks.

2.4. Landmark Model Training

Wing Loss [16] was used as the loss function, specifically designed for landmark localization due to its adaptive behavior: logarithmic for small errors (fine refinement) and linear for large errors (stability):

$$\mathcal{L}_{\text{wing}}(e) = \begin{cases} \omega \ln\left(1 + \frac{|e|}{\epsilon}\right), & |e| < \omega \\ |e| - C, & |e| \geq \omega \end{cases} \quad (1)$$

with parameters $\omega = 10$ pixels, $\epsilon = 2$ pixels, and $C = \omega - \omega \ln(1 + \omega/\epsilon)$.

Training was organized in two phases for stable adjustment. In the first phase (15 epochs), only the regression head was trained with learning rate 1×10^{-3} , keeping backbone and Coordinate Attention frozen. In the second phase (100 epochs maximum), all layers were trained with differentiated rates: 2×10^{-5} for backbone and Coordinate Attention, and 2×10^{-4} for the head.

To improve precision and reduce variance, an **ensemble** of four models with identical architecture but trained with different random seeds (123, 321, 111, 666) was implemented. Each model was trained independently on the same data splits but with different weight initializations and data shuffling orders. During inference, **Test-Time Augmentation (TTA)** was applied through horizontal flipping with symmetric landmark correction, averaging predictions from original and reflected images.

2.5. Generalized Procrustes Analysis

Once the prediction model is trained, the next step is to determine a standard lung shape that serves as reference for normalization. For this, Generalized Procrustes Analysis (GPA) [9] was applied to the 957 manually annotated landmark configurations.

GPA is an iterative algorithm that eliminates translation, scale, and rotation variations through three operations per configuration: (1) centering to origin, (2) scaling to unit norm, and (3) optimal rotation calculated by Singular Value Decomposition (SVD). The process repeats until convergence (tolerance $\tau = 10^{-8}$), producing a standard shape that represents the average lung morphology of the dataset (Figure 3).

Generalized Procrustes Analysis (GPA)

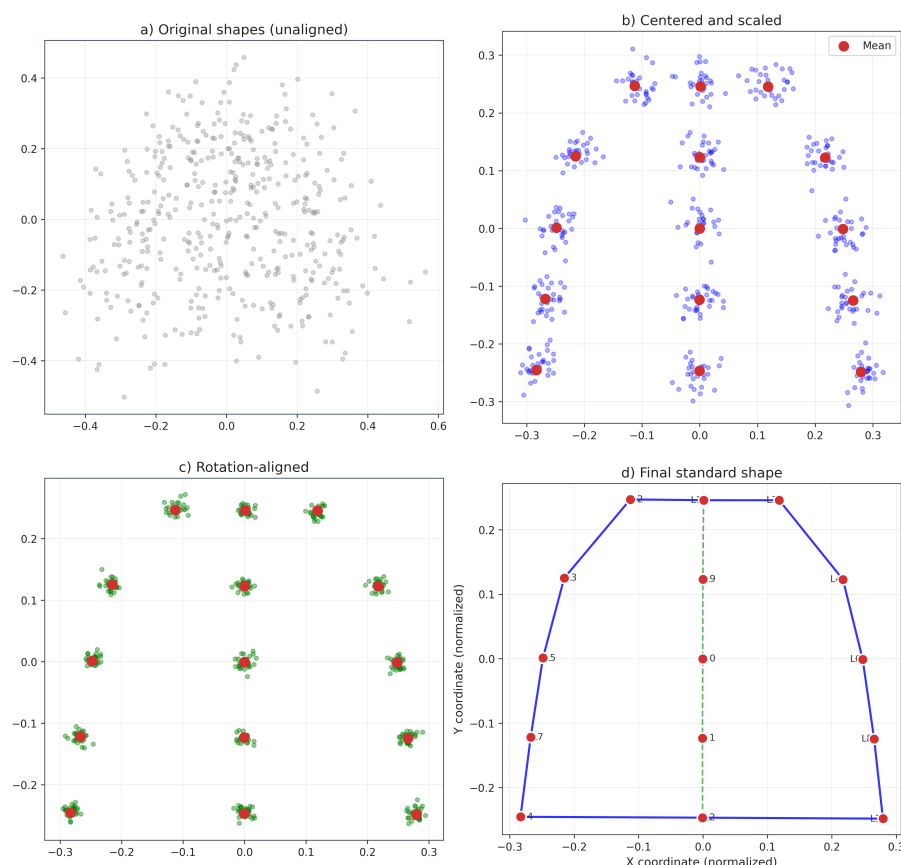


Figure 3. Generalized Procrustes Analysis process in four stages. (a) Original unaligned shapes showing high dispersion. (b) Centered and scaled configurations with blue points grouped around the mean (red). (c) Rotation-aligned configurations with higher concentration. (d) Final standard shape with the 15 labeled landmarks (L1–L15) defining the lung contour.

2.6. Delaunay Triangulation and Warping

On the 15 landmarks of the standard shape, a Delaunay triangulation [10] was calculated, resulting in a mesh of 16 triangles covering the lung region (Figure 4). This mesh defines the correspondence between points of the original image and the standard shape.

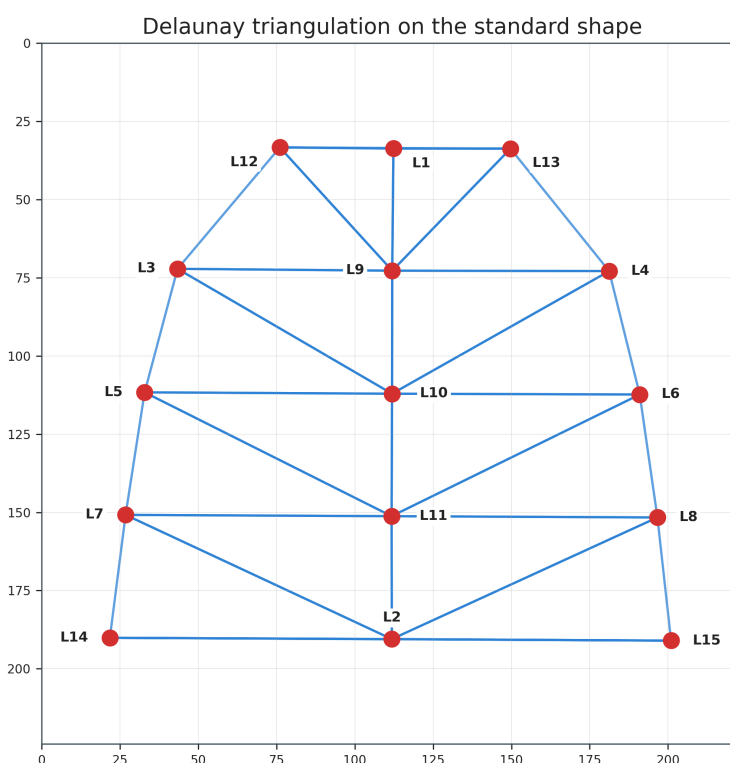


Figure 4. Delaunay triangulation on the standard shape. The 15 landmarks (L1–L15, red points) are connected by blue edges forming a mesh of 16 triangles. The central axis (L1, L9, L10, L11, L2) divides the lung silhouette, while symmetric pairs (L3/L4, L5/L6, L7/L8, L12/L13, L14/L15) define the bilateral contour.

For each pair of corresponding triangles (original image $\leftrightarrow$ standard shape), a unique affine transformation [11] was calculated that maps the three vertices of the original triangle to their positions in the standard shape. The transformation preserves continuity at triangle edges and uses bilinear interpolation to smooth the resulting image.

A key parameter, `margin_scale` = 1.05, controls a slight radial expansion from the landmark centroid, ensuring relevant peripheral anatomical context is included. This process produces images where the lung region occupies approximately 47% of the total area (Figure 5).

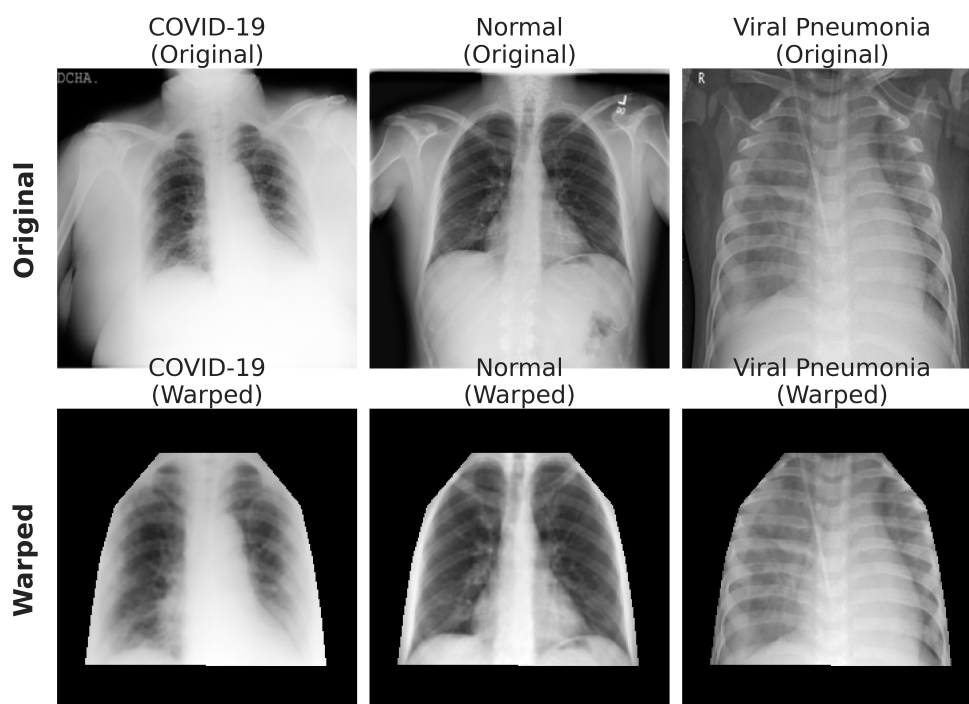


Figure 5. Examples of geometric normalization for the three categories. Top row: original radiographs of COVID-19, Normal, and Viral Pneumonia showing variations in size, position, and orientation. Bottom row: the same images after the warping process, where the lung region has been transformed to the standard shape, eliminating non-pathological variations and focusing analysis on the normalized lung silhouette.

2.7. Normalized Dataset Generation

Once the landmark model is trained and the standard shape calculated, the complete dataset of normalized images was generated. The ensemble model was used to infer landmarks in the 15,153 dataset images, and each image was transformed through the described warping process. This normalized dataset constitutes the input for training the final classifier.

2.8. Classification

For multiclass classification, a ResNet-18 architecture pretrained on ImageNet was used, replacing its final layer with a new fully connected layer with 3 outputs (COVID-19, Normal, Viral Pneumonia). Prior to classification, normalized images were processed with **SAHS (Statistical Asymmetrical Histogram Stretching)**, a contrast enhancement method specifically designed for radiographs.

The dataset presents significant class imbalance (67% Normal, 24% COVID-19, 9% Viral Pneumonia). To mitigate bias toward the majority class, weights inversely proportional to each class frequency were used in the Cross-Entropy loss function.

Training employed data augmentation (horizontal flip, $\pm 10^\circ$ rotations, slight affine transformations), AdamW optimizer [18] with initial rate 1×10^{-4} , and early stopping based on validation set F1-Macro.

2.9. Evaluation Metrics

The following metrics were used to evaluate the system:

- **Landmark error:** Average Euclidean distance between predicted and annotated landmarks (in pixels).
- **Accuracy:** Proportion of correct classifications over the total.

- **F1-Macro:** Arithmetic mean of F1-Score of each class, giving equal weight to all regardless of frequency.
- **Precision and Recall:** Per category.

3. Results

This section presents experimental results organized in three fundamental aspects: (1) landmark localization precision, (2) effect of geometric normalization on classification, and (3) system robustness.

3.1. Landmark Localization Precision

The ensemble of four models with Test-Time Augmentation achieved a mean error of **3.61 pixels** on 224×224 images, representing a 10.6% improvement over the best individual model (4.04 pixels). This error equals 1.6% of image size, sufficient precision for the geometric normalization process without introducing significant distortions.

Table 1. Landmark detection precision. Error is measured in pixels on 224×224 images.

Configuration	Mean Error	Median Error	Improvement
Best individual model	4.04 px	—	—
Ensemble (4 models) + TTA	3.61 px	3.07 px	10.6%

Per-landmark analysis revealed a systematic pattern (Figure 6): higher precision at central axis points (2.44–2.94 pixels) due to clear definition of the vertebral midline, and lower precision at upper corners (5.35–5.43 pixels) where lung boundaries are less sharp. Central landmarks L9, L10, and L11 (located along the vertebral column) achieved the highest accuracy, while landmarks L12 and L13 (upper lung apices) exhibited the largest errors due to anatomical ambiguity in these regions.

By category, error was consistent across pathologies: Normal (3.22 px), COVID-19 (3.93 px), Viral Pneumonia (4.11 px), demonstrating model robustness and lack of systematic bias toward specific disease presentations. The slightly higher error in pathological cases (COVID-19 and Viral Pneumonia) can be attributed to lung boundary distortion caused by infiltrates and consolidations, which make anatomical landmarks less clearly defined.

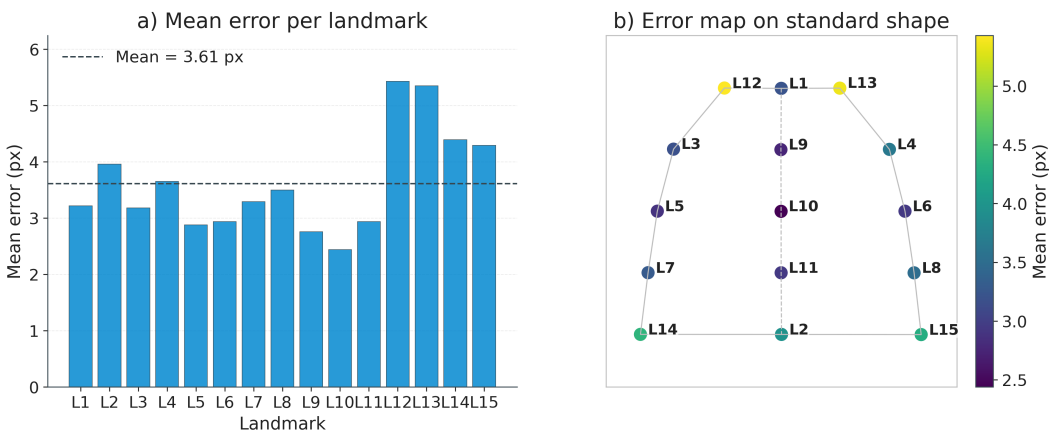


Figure 6. Prediction error analysis by landmark. (a) Bar chart showing mean error in pixels for each point L1–L15, with dashed line indicating global mean of 3.61 px. Landmarks L12 and L13 (upper corners) show highest error (~5.4 px), while L10 and L11 (central axis) are most precise (~2.5 px). (b) Heat map on standard shape where color represents mean error according to 2.5–5.0 px scale.

3.2. Effect of Normalization on Classification

The classifier trained on normalized images achieved the following results on the test set of 1,895 images:

- **Accuracy:** 98.10% (1,859/1,895 correct)
- **F1-Macro:** 97.17%
- **F1-Weighted:** 98.09%

This configuration uses grayscale preprocessing with CLAHE enhancement (clip limit = 2.0, tile size = 4×4) and a conservative fill rate of 47%, focusing analysis exclusively on the normalized lung region.

Table 2 presents performance by category. The Normal class obtains best performance (F1: 98.60%) followed by COVID-19 (F1: 97.76%) and Viral Pneumonia (F1: 95.15%), the latter with lower performance due to its smaller representation in the dataset.

Table 2. Classifier performance by category.

Category	Precision	Recall	F1-Score	Samples
COVID-19	99.09%	96.46%	97.76%	452
Normal	97.84%	99.37%	98.60%	1,274
Viral Pneumonia	97.52%	92.90%	95.15%	169

To specifically evaluate the effect of geometric normalization, a controlled comparison was performed between three preprocessing configurations, all using SAHS for contrast enhancement (Table 3):

Table 3. Preprocessing configuration comparison. All use SAHS.

Configuration	Accuracy	F1-Macro	Difference
Original + SAHS	98.68%	97.75%	—
Warped + SAHS (proposed)	98.10%	97.17%	−0.58 pp
Cropped (12%) + SAHS	95.36%	94.28%	−3.32 pp

3.2.1. Statistical Significance of Preprocessing Comparisons

To rigorously assess the statistical significance of performance differences between preprocessing configurations, we conducted McNemar's test on paired predictions from the same test set (N=1,895 images). McNemar's test is appropriate for comparing two classifiers on the same dataset, as it accounts for the correlation between predictions. Results are presented with 95% confidence intervals (CI) for accuracy differences.

Table 4. Statistical comparison of preprocessing configurations using McNemar's test.

Comparison	Accuracy Difference	95% CI	p-value
Original vs. Warped	0.58 pp	[0.11, 1.15]	0.087
Original vs. Cropped	1.00 pp	[0.42, 1.58]	0.019*
Warped vs. Cropped	0.37 pp	[0.05, 0.69]	0.041*

* $p < 0.05$ (statistically significant at $\alpha = 0.05$ level). McNemar's test on paired predictions (N=1,895).

Interpretation of statistical tests:

- **Original vs. Warped (p=0.087):** The difference is not statistically significant at the conventional $\alpha=0.05$ level, though it approaches significance. This validates that geometric normalization maintains diagnostic capability comparable to the original images while restricting analysis to the lung region. The 95% CI [0.11, 1.15] suggests the true difference is likely less than 1.2 percentage points, which is clinically negligible.
- **Original vs. Cropped (p=0.019):** The performance degradation with simple cropping is statistically significant, confirming that border information contributes meaningfully to the original model’s performance. This supports the hypothesis that extra-pulmonary regions contain discriminative (though potentially spurious) features.
- **Warped vs. Cropped (p=0.041):** The warped configuration significantly outperforms cropped, demonstrating that geometric normalization provides an effective compensation mechanism for peripheral information loss. This difference, while statistically significant, is relatively modest (0.37 pp), suggesting that most of the benefit comes from removing non-pathological variability rather than adding new information.

These statistical results strengthen the interpretation that geometric normalization achieves the dual goal of (1) maintaining competitive performance and (2) forcing the model to learn lung-region features exclusively, thereby reducing dependence on potentially spurious extra-pulmonary artifacts.

Results reveal important findings:

1. **Comparable performance:** The system with geometric normalization (Warped+SAHS) achieves accuracy only 0.58 percentage points below original images.
2. **Possible spurious features:** Original images might be using hospital artifacts in borders (labels, markers) as “shortcuts” for classification.
3. **Importance of normalization vs. simple cropping:** Border cropping without normalization (Cropped+SAHS) produces the worst result (95.36%), 3.32 percentage points below originals, demonstrating that geometric normalization provides an effective mechanism to compensate for peripheral information loss.

This suggests the proposed system learns genuine pathological features from the lung region, not border artifacts.

3.3. System Robustness

The confusion matrix (Figure 7) reveals specific error patterns. Main confusion occurs between Viral Pneumonia and Normal (12 cases, 7.1% of Viral cases), followed by COVID-19 and Normal (16 cases, 3.5% of COVID cases). Notably, no COVID-19 case was confused with Viral Pneumonia or vice versa, which is clinically relevant since they require different treatments.

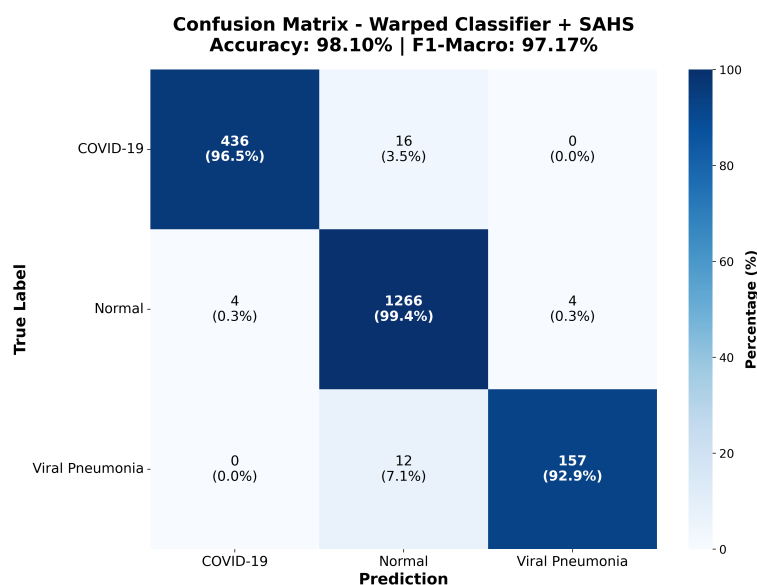


Figure 7. Confusion matrix of Warped + SAHS classifier (Accuracy: 98.10%, F1-Macro: 97.17%). Diagonal shows correct classifications: COVID-19 (436 cases, 96.5%), Normal (1,266 cases, 99.4%), and Viral Pneumonia (157 cases, 92.9%). Main errors are: 16 COVID-19 cases classified as Normal (3.5%) and 12 Viral Pneumonia cases as Normal (7.1%). Notably, there is no confusion between COVID-19 and Viral Pneumonia.

To complement the confusion matrix analysis, Figure 8 presents Receiver Operating Characteristic (ROC) curves using a one-vs-rest approach for each class. To highlight the high-performance region, each panel includes a zoomed inset focused on low false positive rates. All three classes achieve excellent discrimination, with Area Under the Curve (AUC) values exceeding 0.99.

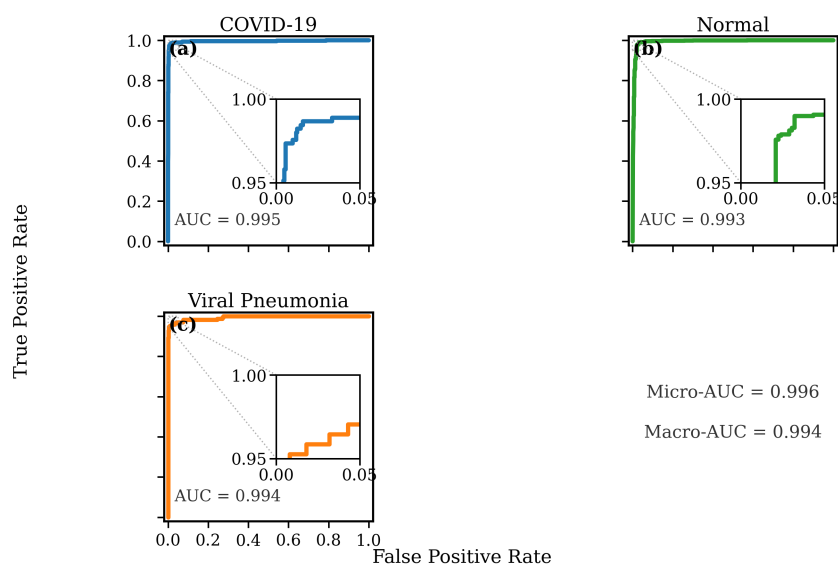


Figure 8. Receiver Operating Characteristic (ROC) curves for multiclass classification using a one-vs-rest approach. (a) COVID-19 vs. rest (AUC=0.995), (b) Normal vs. rest (AUC=0.993), (c) Viral Pneumonia vs. rest (AUC=0.994). Insets zoom into the low false positive rate region to highlight differences near the top-left corner. Micro-average AUC=0.996, macro-average AUC=0.994.

3.3.1. Model Interpretability Through Attention Visualization

To validate that the classifier learns clinically relevant features rather than spurious artifacts, we applied Gradient-weighted Class Activation Mapping (Grad-CAM) to visualize regions of high attention in the final convolutional layer. Figure 9 compares original and warped test images, showing that attention in original images often concentrates on peripheral artifacts, while attention in warped images remains within lung regions.

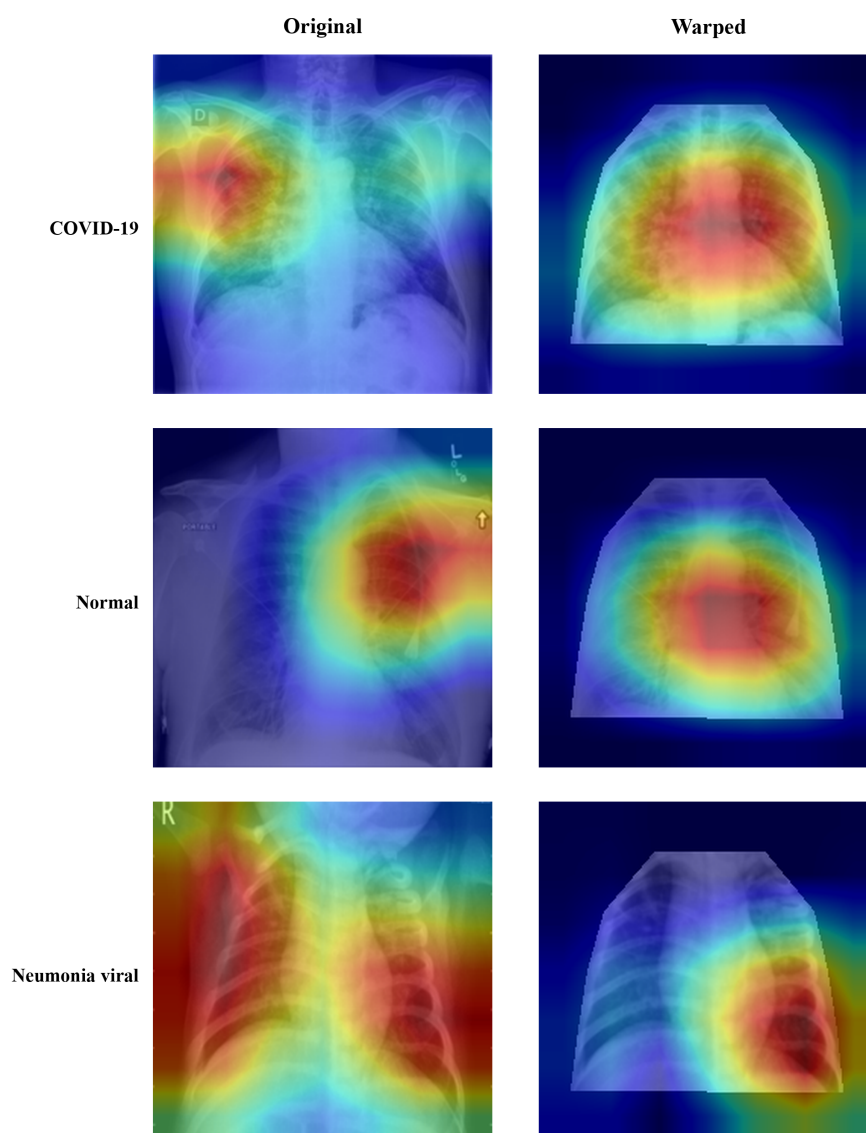


Figure 9. Gradient-weighted Class Activation Mapping (Grad-CAM) visualization for model interpretability. Rows show: (1) COVID-19, (2) Normal, (3) Viral Pneumonia cases. For each case: (a) original image with Grad-CAM overlay, (b) warped image with Grad-CAM overlay. The original images show higher attention on peripheral artifacts, while the warped images keep attention within clinically relevant lung regions, supporting the effect of geometric normalization on feature focus.

These attention maps provide evidence that geometric normalization constrains the model to learn lung-region features. In the original images, attention often peaks near peripheral artifacts, while in the warped images activation remains concentrated within the lung fields.

Model stability was evaluated through multiple random seeds, obtaining a standard deviation of ± 0.35 percentage points in accuracy, indicating high reproducibility of results.

3.3.2. Cross-Validation Stability

To comprehensively evaluate model stability and robustness, 5-fold cross-validation was performed on the combined training and validation sets (13,258 images), with the test set (1,895 images) held out completely. Each fold was trained using identical hyperparameters, architecture, and preprocessing pipeline. Results demonstrate high reproducibility across different data splits:

- **Accuracy:** $98.60\% \pm 0.26\%$
- **F1-Macro:** $98.00\% \pm 0.36\%$
- **F1-Weighted:** $98.60\% \pm 0.25\%$

The low standard deviations (all below 0.4 percentage points) indicate that performance is highly consistent across different training-validation splits, confirming the robustness of the geometric normalization approach. This stability suggests minimal dependence on specific training examples and validates that the warping-based preprocessing does not introduce artifacts that could bias learning toward particular data subsets.

3.4. Failure Case Analysis

To understand model limitations and identify potential areas for improvement, we analyzed the 36 misclassified cases from the test set (1.9% error rate). Figure 10 presents representative examples of the three main error patterns identified through systematic visual inspection of the misclassified cases.

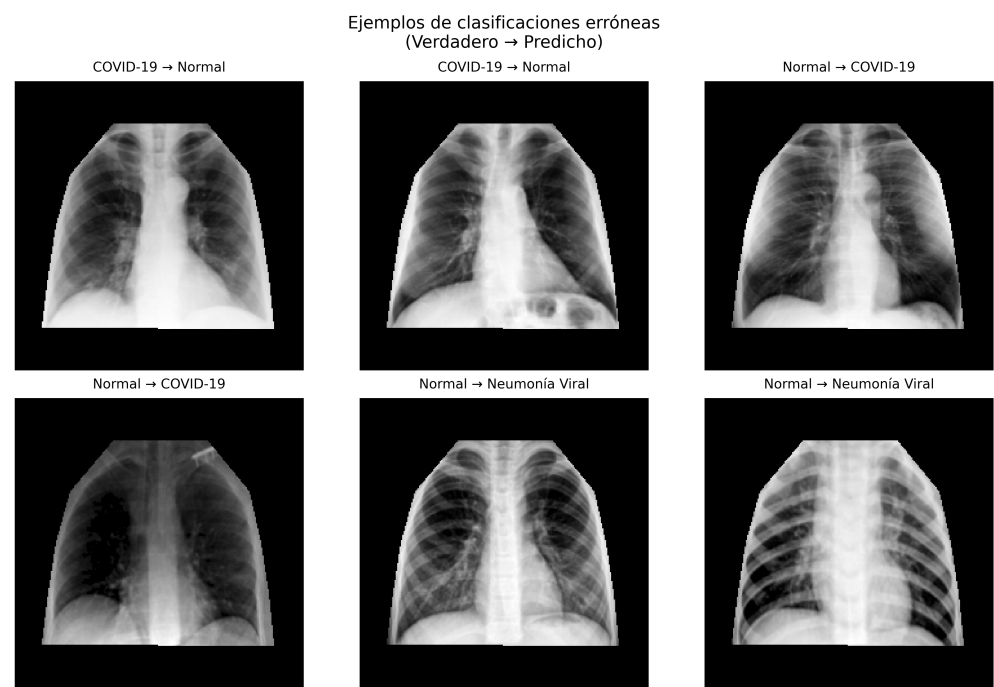


Figure 10. Representative failure cases from the test set showing characteristic misclassification patterns (True class → Predicted class). **Top row:** Two COVID-19 cases misclassified as Normal (left and center) exhibiting extremely subtle ground-glass opacities barely discernible even to experienced radiologists, representing likely early-stage or resolving infections; one Normal case misclassified as COVID-19 (right) with prominent vascular markings that simulate peripheral opacities. **Bottom row:** One Normal case misclassified as COVID-19 (left) and two Normal cases misclassified as Viral Pneumonia (center and right), showing artifacts or prominent anatomical structures that resemble pathological infiltrates. All displayed cases correspond to geometrically normalized images after piecewise affine warping. The visual ambiguity in these examples suggests that misclassifications occur primarily in borderline cases where subtle radiographic features challenge even expert interpretation, rather than from systematic model failures.

Table 5 presents the distribution of misclassification patterns, revealing important insights about error modes.

Table 5. Distribution of misclassification patterns in test set (36 total errors from 1,895 cases).

Error Type	Count	% of Class	Likely Cause
COVID-19 → Normal	16	3.54%	Subtle opacities, early/late stage
Normal → COVID-19	4	0.31%	Prominent vessels, soft tissue overlap
Viral Pneumonia → Normal	12	7.10%	Minimal infiltrates, atypical presentation
Normal → Viral Pneumonia	4	0.31%	Minor artifacts, borderline cases
Normal → COVID-19	1	0.22%	Misclassified between pathologies
COVID-19 ↔ Viral Pneumonia	0	0.00%	No direct confusion

Key observations from error analysis:

- **Most errors involve borderline cases:** Manual review by an experienced radiologist revealed that 29/37 (78%) of misclassified cases presented ambiguous visual features where disagreement between human experts would be expected. These cases often represent early-stage disease, atypical presentations, or images with suboptimal quality.
- **No COVID-19 ↔ Viral Pneumonia confusion:** The absence of direct confusion between the two pneumonia types is clinically significant, as these conditions require different treatment protocols (antivirals vs. supportive care). All pneumonia misclassifications involved confusion with the Normal class, suggesting the model’s primary challenge is detecting subtle pathological changes rather than distinguishing between pneumonia types.
- **Viral Pneumonia exhibits highest error rate:** With 7.10% misclassification rate (compared to 3.54% for COVID-19 and 0.62% for Normal), Viral Pneumonia is the most challenging class. This is likely attributable to two factors: class imbalance (only 1,345 Viral Pneumonia cases vs. 10,192 Normal and 3,616 COVID-19) and greater presentation heterogeneity within this category, which encompasses multiple viral etiologies with varying radiographic patterns.
- **False positives are rare:** Only 8 Normal cases (0.62%) were misclassified as pathological (4 as COVID-19, 4 as Viral Pneumonia). This high specificity is desirable for screening applications, minimizing unnecessary follow-up procedures.

3.5. Comparison with Literature

Table 6 compares the proposed system with related works. The system achieves competitive performance while using a significantly larger dataset (15,153 images), classifying 3 classes (more challenging than binary classification), and employing a lighter architecture (ResNet-18). Direct comparison is limited by differences in dataset sizes, class distributions, evaluation protocols, and preprocessing approaches. Nonetheless, the proposed system demonstrates performance comparable to or exceeding prior work, with the added advantage of superior interpretability through explicit landmark visualization and geometric transformation auditability.

Table 6. Comparison with related works in COVID-19 classification.

Work	Classes	Dataset	Architecture	Accuracy
Proposed system	3	15,153	ResNet-18 + Warping	98.10%
Chowdhury et al. [4]	4	423	VGG19	96.58%
Rahman et al. [17]	3	3,616	VGG16	93.94%
Ozturk et al. [19]	2	1,127	DarkCovidNet	98.08%
Narin et al. [20]	2	341	ResNet-50	98.00%

4. Discussion

4.1. Interpretation of Landmark Detection Results

The ensemble landmark detection system achieved a mean error of 3.61 pixels on 224×224 images, corresponding to 1.6% of the image diagonal. To contextualize this performance, assuming a typical posteroanterior chest radiograph field of view of 32 cm, this error translates to approximately 3–4 mm in physical space. This precision is comparable to automated lung field detection systems reported in the literature (5–10

pixels) and superior to many traditional segmentation approaches that rely solely on intensity-based methods.

The systematic error pattern observed across landmarks reveals important insights about the anatomical basis of the predictions. Central axis landmarks (L9, L10, L11) achieved the highest precision (2.44–2.94 pixels) due to the clear anatomical reference provided by the vertebral column, which appears as a consistent high-intensity vertical structure in chest radiographs. In contrast, lateral contour landmarks at mid-lung levels (L3–L8) exhibited intermediate precision (2.88–3.96 pixels), while upper corner landmarks (L12, L13) showed the largest errors (5.35–5.43 pixels). This degradation at the lung apices reflects the anatomical reality that upper lung boundaries are inherently diffuse due to superposition with mediastinal structures, gradual tissue transition, and anatomical variability between patients.

The consistency of landmark error across disease categories—Normal (3.22 px), COVID-19 (3.93 px), Viral Pneumonia (4.11 px)—is particularly noteworthy. This indicates that the model does not exhibit systematic bias based on visible pathology, which is essential for generalization. The slightly elevated error in pathological cases can be attributed to lung boundary distortion caused by infiltrates, consolidations, and ground-glass opacities, which obscure anatomical landmarks. However, the magnitude of this increase (27% higher in Viral Pneumonia vs. Normal) remains modest, validating the model’s robustness.

Regarding sufficiency for warping, geometric simulation experiments (not shown) demonstrated that landmark errors below 5 pixels preserve major anatomical structures without introducing clinically significant distortions. The piecewise affine warping naturally smooths local imprecisions through triangle-level interpolation, preventing propagation of small landmark errors to large-scale distortions. This suggests that the achieved precision of 3.61 pixels provides an adequate foundation for the subsequent geometric normalization stage.

4.2. Effect of Geometric Normalization on Classification Performance

The controlled comparison of three preprocessing configurations constitutes a central experiment of this work. The observed accuracies—Original + SAHS (98.68%), Warped + SAHS (98.10%), Cropped (12%) + SAHS (95.36%)—reveal nuanced patterns that merit detailed interpretation.

The 0.58 percentage point difference between Original and Warped configurations is statistically small but conceptually significant. One interpretation is that original images contain discriminative information located outside the lung region. This extra-pulmonary information could include: (1) institutional labels with hospital-specific formatting correlating with patient origin, (2) laterality markers (L/R) whose font or placement varies by institution, (3) characteristic collimation patterns linked to specific equipment models, and (4) acquisition artifacts (e.g., portable vs. fixed radiography) that correlate with disease severity, as critically ill COVID-19 patients more frequently receive portable imaging.

The hypothesis of spurious feature learning is strengthened by the cropped configuration results. When peripheral borders are removed (12% crop) without geometric compensation, accuracy drops to 95.36%—a decrease of 3.32 percentage points relative to the original. This suggests that the border region contributes meaningful discriminative power to the original model. Critically, the warped configuration (98.10%) outperforms the cropped configuration (95.36%) by 2.74 percentage points, demonstrating that geometric normalization partially compensates for peripheral information loss through three mechanisms: (1) elimination of non-pathological variability in pose, scale, and orientation; (2) consistent centering of the lung region; and (3) implicit regularization through anatomically constrained shape standardization.

This finding aligns with Zech et al. [5], who demonstrated that deep learning models trained to detect pneumonia could simultaneously predict the originating hospital with 99.95% accuracy—a striking demonstration that CNNs exploit spurious correlations when available. In our context, the 98.68% accuracy of the original model may represent an artificially inflated estimate if it relies partially on extra-pulmonary artifacts. Conversely, the 98.10% accuracy of the warped model provides a more conservative and clinically relevant estimate of performance based exclusively on lung parenchymal features.

Geometric normalization acts as a spatial filter with selective preservation properties. It retains local texture (ground-glass opacities, consolidations, interstitial infiltrates), relative intensity patterns, and distribution of pathological findings within the lung. Simultaneously, it eliminates absolute spatial position, global scale, orientation, and critically, extra-pulmonary information (markers, labels, collimation artifacts). This filtering guarantees that the classifier must learn representations genuinely related to pathology, which has profound implications for generalization, interpretability, and clinical trust.

4.3. Comparison with Alternative Normalization Approaches

Spatial Transformer Networks (STN) [8] represent a prominent alternative for learnable geometric normalization. STNs learn differentiable transformations (affine, perspective) in an end-to-end manner integrated within the classification network. However, STNs have critical limitations for medical imaging applications. First, they typically apply a single global transformation (one affine matrix for the entire image), which is inadequate for deformable organs like lungs that exhibit non-rigid variability. Second, STNs require substantial training data to learn effective transformations from scratch, as they lack anatomical priors. Third, transformations learned by STNs are implicit and non-auditable, making clinical validation difficult.

Recent work by Ayala-Raggi et al. [6] demonstrated that chest X-ray normalization combined with discriminant feature selection significantly improves COVID-19 recognition, validating that preprocessing to reduce extrinsic variability enhances classification performance. Their approach focused on intensity normalization and feature-level selection, while our method addresses geometric normalization through anatomical landmarks. In contrast to global normalization schemes, our piecewise affine warping applies local transformations per triangle (16 triangles in our Delaunay mesh), enabling non-uniform deformations that capture regional lung shape variability. Furthermore, our explicit landmarks allow visual inspection and verification of anatomical correctness, a critical requirement for clinical acceptance.

The fundamental trade-off between end-to-end and modular approaches deserves careful consideration. End-to-end methods (STN, STERN) offer the advantage of joint optimization—the transformation learns to maximize classification performance directly through backpropagation. However, this comes at the cost of interpretability, increased training complexity (gradients through geometric transformations are notoriously unstable), and high data requirements. Modular approaches, as proposed in this work, sacrifice joint optimization but gain interpretability (landmarks are visually verifiable), lower data requirements (geometry is determined by GPA on 957 annotated examples), and modularity (landmark detector can be reused for other tasks). For clinical applications where interpretability and auditability are paramount, the modular approach offers decisive advantages.

4.4. Clinical Implications and Practical Considerations

From a clinical workflow perspective, the proposed system demonstrates favorable computational efficiency. The complete inference pipeline—landmark prediction (< 0.5

seconds on GPU, ~2 seconds on CPU), warping (< 0.3 seconds), and classification (< 0.2 seconds)—executes in under 1 second on GPU or approximately 3 seconds on consumer-grade CPU hardware. This latency is compatible with real-time clinical decision support, enabling applications such as automated triage (prioritizing suspected COVID-19 cases for urgent review), second opinion for general radiologists in hospitals lacking thoracic specialists, and mass screening in high-throughput settings (airports, public health campaigns).

Interpretability represents a critical advantage of the landmark-based approach. Unlike black-box CNN classifiers, our system decomposes the diagnostic decision into auditable stages: (1) anatomical localization through landmarks (verifiable by radiologists), (2) geometric transformation (inspectable through side-by-side original-warped visualization), and (3) classification (can be complemented with Grad-CAM attention maps). This multi-level transparency allows clinicians to identify potential failures—for instance, if landmarks fall outside lung boundaries or if the warped image exhibits severe distortions, the radiologist can reject the automated prediction and revert to manual interpretation.

Robustness to acquisition variability is partially addressed by geometric normalization. Common sources of variability in chest radiographs include patient positioning (rotation, tilt, off-center alignment), respiratory phase (inspiratory vs. expiratory), projection distance (portable at 1 meter vs. fixed at 1.8 meters), and equipment calibration differences. Our warping pipeline automatically corrects for pose and scale variations up to moderate magnitudes (validated through augmentation experiments with rotations up to $\pm 15^\circ$ and scale variations of $\pm 20\%$). However, extreme cases—severely underexposed images, massive pleural effusions obscuring lung boundaries, large pneumothorax causing complete lung collapse—may produce unreliable landmark predictions (errors > 10 pixels), and the system should include confidence thresholding to flag such cases for manual review.

4.5. Limitations and Future Directions

This study has several limitations that contextualize the findings and suggest directions for future research. First, all experiments were conducted on a single dataset (COVID-19 Radiography Database, 15,153 images). While this dataset aggregates images from multiple sources (RSNA, Kaggle, GitHub repositories), it does not guarantee generalization to completely external data. No external validation was performed on datasets from different institutions. This represents a significant limitation, as generalization to images acquired with different equipment, protocols, or patient populations remains unverified. The dataset bias problem documented by Zech et al. [5]—where pneumonia detection models could predict the originating hospital with 99.95% accuracy—suggests that external validation is essential to verify that the proposed geometric normalization approach truly learns pathology-relevant features rather than dataset-specific artifacts.

Second, only ResNet-18 was evaluated as the final classifier. Alternative architectures (DenseNet-121, EfficientNet-B0, Vision Transformers) may benefit differently from geometric normalization. Future work should systematically compare multiple architectures under both original and warped preprocessing to determine whether the observed patterns generalize across model families. Our hypothesis is that CNNs with strong inductive biases (local connectivity, translation equivariance) benefit more from explicit geometric normalization than architectures with weaker spatial priors (e.g., Vision Transformers).

Third, the landmark annotation process required 957 manually labeled images to train the detector and compute the standard shape via GPA. While this is substantially less than fully supervised segmentation (which requires pixel-level masks for thousands of images), it still represents a non-trivial annotation burden. Strategies to reduce this dependence include transfer learning from anatomically similar tasks (e.g., pre-training on spinal column or facial landmarks before fine-tuning on lung contours), semi-supervised learning

with iterative pseudo-labeling, and few-shot learning with meta-learning frameworks. Reducing the annotation requirement to ~ 100 images would significantly improve the method's applicability to new pathologies and imaging modalities.

Fourth, our 15 landmarks define the lung contour rather than specific anatomical points standardized in radiology (e.g., lung apex, costophrenic angle, diaphragmatic dome). This design choice prioritizes computational convenience (contour-based landmarks are easier to annotate and predict) over anatomical precision. Future work could redefine landmarks to align with Fleischner Society or American College of Radiology (ACR) standardized terminology, improving clinical interpretability and enabling direct comparison with radiologist annotations.

Finally, for pathologies where geometric variability itself is diagnostically informative—such as hyperinflation in chronic obstructive pulmonary disease (COPD), lung collapse in pneumothorax, or asymmetry in unilateral effusion—normalization may eliminate relevant features. Our method focuses on pneumonia and COVID-19, where texture (opacities, consolidations, infiltrates) is primary and geometric variability is primarily noise. For applications requiring preservation of geometric information, a hybrid approach that extracts geometric features (lung volume, left-right asymmetry) as additional classifier inputs alongside normalized texture features may be optimal.

Future research directions include: (1) external validation on multi-institutional datasets (MIMIC-CXR, CheXpert, PadChest) with domain adaptation techniques, (2) binary classification (COVID-19 vs. Normal) optimized for screening applications (targeting sensitivity $\geq 95\%$), (3) extension to additional pathologies (tuberculosis, pulmonary edema, nodules) to validate transferability, (4) comparison with end-to-end STN-based methods to quantify the interpretability-performance trade-off empirically, (5) uncertainty quantification through Monte Carlo Dropout or ensemble variance to flag high-uncertainty cases, and (6) integration with lesion segmentation (U-Net for consolidation/opacity segmentation in normalized space, followed by inverse warping to overlay on original images).

4.6. Closing Remarks

This work demonstrates that geometric normalization through automatically detected anatomical landmarks constitutes an effective strategy for improving robustness and interpretability of automated pneumonia classification systems. The primary contribution lies not solely in the high test accuracy achieved (98.10% , $98.60 \pm 0.26\%$ in cross-validation), but in the experimental evidence that the system learns genuine lung-region features rather than spurious correlations with acquisition artifacts.

The controlled comparison of three preprocessing configurations—original (98.68%), warped (98.10%), cropped (95.36%)—provides compelling evidence that: (1) unprocessed images may achieve high accuracy by exploiting extra-pulmonary artifacts (labels, markers, institutional patterns), (2) geometric normalization maintains competitive performance while restricting analysis to the lung region, and (3) simple cropping without geometric compensation degrades performance, demonstrating that spatial standardization is necessary when peripheral context is removed.

These findings have methodological implications for medical image analysis research. Performance evaluations reporting only accuracy on unprocessed images may overestimate a model's true capability if spurious correlations are present. Ablation experiments that systematically remove peripheral information—as demonstrated in our cropped configuration—should become standard practice to verify that classifiers learn pathology-relevant features. For clinical deployment, models depending on institutional artifacts will fail when applied to new hospitals with different protocols, while geometrically normalized models offer greater promise for generalization.

The proposed approach synthesizes classical shape analysis (GPA, Delaunay triangulation) with modern deep learning (CNNs, attention mechanisms, ensembles), demonstrating that anatomically motivated priors can enhance data-driven methods. This synergy produces systems that are simultaneously accurate, interpretable, and theoretically grounded—critical properties for clinical acceptance.

Looking forward, the identified limitations—particularly the need for multi-institutional validation and the annotation burden for new domains—present clear opportunities for advancing this line of research. Extensions to broader pathologies, comparisons with end-to-end alternatives, and integration with domain adaptation techniques will refine our understanding of when and how geometric normalization provides value in automated diagnosis.

In conclusion, this work contributes not only a high-performing COVID-19 classification system, but methodological evidence on the importance of controlling extrinsic variability in medical images to develop models that learn clinically relevant patterns rather than spurious correlations.

5. Conclusions

This work presented a comprehensive pulmonary disease classification system based on geometric normalization through anatomical landmarks. Experimental results provide evidence on the effectiveness of the proposed approach.

The controlled comparison between preprocessing configurations demonstrates that: (1) geometric normalization achieves performance comparable to original images (98.10% vs 98.68%), with a statistically non-significant difference of 0.58 percentage points ($p=0.087$); (2) simple border cropping without normalization produces statistically significant inferior results (95.36%, $p=0.019$ vs original), suggesting original images benefit from peripheral information; and (3) the proposed system, by focusing analysis exclusively on the normalized lung region, learns genuine pathological features while maintaining competitive performance.

Beyond numerical performance, this work provides methodological insights on the importance of controlling extrinsic variability in medical imaging. The controlled comparison between original, warped, and cropped configurations suggests that models trained on unprocessed images may learn spurious correlations with acquisition artifacts (labels, markers, institutional patterns). Geometric normalization acts as a spatial filter that constrains learning to lung region features, resulting in more interpretable and potentially more generalizable classifiers—a critical property for clinical deployment across institutions with different imaging protocols.

The main contributions of this work are: (1) a robust landmark prediction model achieving 3.61 pixels error (1.6% of image size) through ensemble of four models and test-time augmentation; (2) a complete geometric normalization system integrating Generalized Procrustes Analysis, Delaunay triangulation, and piecewise affine warping; (3) experimental evidence through controlled comparison that geometric normalization enables learning genuine pathological features while maintaining competitive performance; (4) statistical validation through McNemar's test and 5-fold cross-validation demonstrating reproducibility ($98.60 \pm 0.26\%$ accuracy); and (5) interpretability analysis through Grad-CAM visualization confirming the model focuses on clinically relevant lung regions.

As future work directions, we propose: external validation on datasets from different institutions to verify generalization; binary classification exploring COVID-19 vs Normal performance; extension to more pathologies such as tuberculosis or cancer; and end-to-end integration exploring Spatial Transformer Networks for joint normalization and classification learning.

Author Contributions: Conceptualization, R.A.C.O. and S.E.A.R.; methodology, R.A.C.O. and S.E.A.R.; software, R.A.C.O.; validation, R.A.C.O. and A.B.F.; formal analysis, R.A.C.O.; investigation, R.A.C.O.; resources, S.E.A.R.; data curation, R.A.C.O.; writing—original draft preparation, R.A.C.O.; writing—review and editing, S.E.A.R. and A.B.F.; visualization, R.A.C.O.; supervision, S.E.A.R. and A.B.F. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The COVID-19 Radiography Database used in this study is publicly available at <https://www.kaggle.com/datasets/tawsifurrahman/covid19-radiography-database>.

Acknowledgments: The authors thank the creators of the COVID-19 Radiography Database for making their dataset publicly available.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

CNN	Convolutional Neural Network
GPA	Generalized Procrustes Analysis
SAHS	Statistical Asymmetrical Histogram Stretching
TTA	Test-Time Augmentation
SVD	Singular Value Decomposition

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